



The Chameleon’s Limit: Investigating Persona Collapse and Homogenization in Large Language Models

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Codebase

Abstract

Applications based on large language models (LLMs), such as multi-agent simulations, require population diversity among agents. We identify a pervasive failure mode we term *Persona Collapse*: agents each assigned a distinct profile nonetheless converge into a narrow behavioral mode, producing a homogeneous simulated population. To quantify persona collapse, we propose a framework that measures how much of the persona space a population occupies (Coverage), how evenly agents spread across it (Uniformity), and how rich the resulting behavioral patterns are (Complexity). Evaluating ten LLMs on personality simulation (BFI-44), moral reasoning, and self-introduction, we observe persona collapse along two axes: (1) Dimensions: a model can appear diverse on one axis yet structurally degenerate on another, and (2) Domains: the same model may collapse the most in personality yet be the most diverse in moral reasoning. Furthermore, item-level diagnostics reveal that behavioral variation tracks coarse demographic stereotypes rather than the fine-grained individual differences specified in each persona. Counter-intuitively, **the models achieving the highest per-persona fidelity consistently produce the most stereotyped populations**. We release our toolkit and data to support population-level evaluation of LLMs.

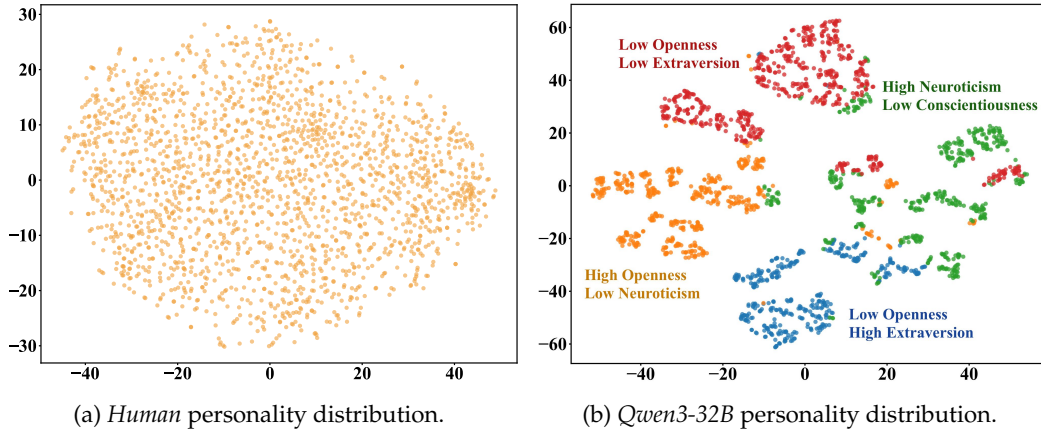


Figure 1: t-SNE projection of the BFI-44 personality instrument for 2,058 individuals. **(a)** Human respondents spread diffusely across the space. **(b)** When given persona prompts, Qwen3-32B responses fragment into separated clusters rather than filling the space.

1 Introduction

Agents powered by large language models (LLMs) are increasingly deployed as participants in simulated societies (Park et al., 2023; Zhou et al., 2025), synthetic survey respondents, and proxies for user testing (Louie et al., 2024). These applications rest on a critical assumption: given a richly specified persona (e.g., age, gender, nationality, political leaning, and occupation), the model can simulate an individual whose behavior reflects the complex intersection of all these attributes (Xiao et al., 2025). However, existing evaluations (Wang et al., 2024; Abdulhai et al., 2025) primarily measure “shallow fidelity”: assessing whether an agent successfully mimics a simple, low-dimensional label, such as a well-known public figure or a broad demographic prototype (e.g., “a sixty-year-old black woman”).

While such shallow fidelity may appear sufficient, it remains unclear whether LLMs actually maintain fidelity or more complicated simulation targets or if they silently ignore information. In this study, we identify a fundamental failure in such “combinatorial fidelity.” When instructed to role-play a persona defined by 26 distinct dimensions, LLMs systematically retain only the most stereotypically salient attributes for downstream tasks, completely discarding the rest. For instance, persona profiles specified as *an older, married, upper-class Muslim woman of Middle Eastern origin residing in the United States* and as *a young, divorced, homosexual, lower-class non-binary individual living in India* converge onto similar outputs on the personality evaluation, driven solely by their shared attributes of *high extraversion and low openness*. This attribute truncation severely degrades the genuine behavioral richness of the simulated population. As illustrated in Fig. 1, a thousand supposedly distinct agents degenerate into several stereotyped clusters, in contrast to the diversity of humans. We term this structural homogenization **Persona Collapse**.

To systematically measure persona collapse, we shift from individual-level fidelity metrics to population-level diagnostics. We formalize this analysis through the *Behavioral Trait Matrix* $\mathbf{B} \in \mathbb{R}^{N \times D}$, where each row represents a single persona’s response across D behavioral items. A structurally healthy simulated population should: (1) span the full distribution of human behavioral archetypes rather than over-sampling a modal region and neglecting the tails (**Coverage**; Naeem et al. (2020)) (2) distribute evenly across the behavioral space rather than collapsing into a few dense, degenerate clusters (**Uniformity**; Hopkins & Skellam (1954); Wang & Isola (2020)) (3) be genuinely high-dimensional rather than compressed onto a low-dimensional subspace (**Complexity**; Levina & Bickel (2004); Tulchinskii et al. (2023)). Together, these three criteria form a diagnostic framework grounded in the geometry of $\mathbf{B}$. Coverage and Uniformity capture whether the population looks spread out, while Complexity captures whether that spread is structurally rich or hollow. Since these population-level metrics do not reveal *where* simulation breaks down, we introduce complementary item-level diagnostics (Tucker, 1951; John & Srivastava, 1999) to localize the underlying mechanics, revealing precisely which personality facets are flattened, which moral judgments are stereotyped, and which demographic attributes drive the truncation.

Applying this framework to ten LLMs across structured personality simulation (BFI-44 (John & Srivastava, 1999)), moral reasoning (Liu et al., 2025), and free-form self-introductions, we use true human responses as a normative baseline to demonstrate that all evaluated models exhibit structural collapse. Specifically, we uncover two empirical findings that define the mechanics of this phenomenon: (1) Models that scores highest on per-persona fidelity consistently produce the most collapsed populations. (2) A model can act as a structurally degenerate personality simulator yet appear as a highly diverse moral reasoner. Our contributions are as follows:

- ① We propose a geometric diagnostic framework to systematically measure persona collapse, quantifying the loss of high-dimensional human attributes in LLM simulations.
- ② We evaluate ten LLMs across personality simulation, moral reasoning, and open-ended self-introduction, revealing pervasive persona collapse across all models.
- ③ We demonstrate that models with the highest fidelity produce the most collapsed profiles, exposing a fundamental tension between role adherence and population diversity.

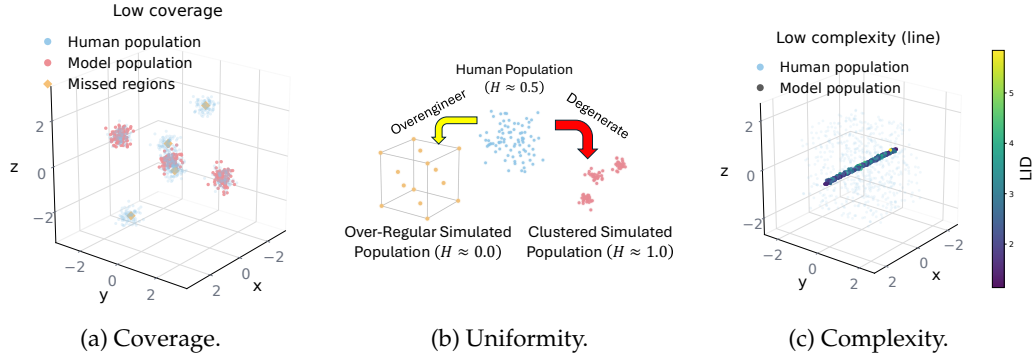


Figure 2: Conceptual illustrations of the three diagnostic axes. **Coverage**: The model concentrates in modal regions. **Uniformity**: Human distributions resemble spatial randomness ($H \approx 0.5$); models either overengineer populations into lattices ($H \rightarrow 0$) or degenerate into isolated clusters ($H \rightarrow 1$). **Complexity**: Humans fill a high-dimensional volume, whereas models collapse onto low-dimensional manifolds (e.g., a line).

2 Diagnosing Population-Level Behavioral Diversity

Personality psychology has long established that human populations exhibit weakly clustered high-dimensional variation between trait instruments (Costa & McCrae, 1992; John & Srivastava, 1999): individuals within the same demographic group differ nearly as much as individuals between groups. A simulated population that claims to represent human diversity should approximate these distributional properties. To test whether it does, we introduce a shared representation and three diagnostic axes.

2.1 The Behavioral Trait Matrix

We represent a population of N personas as a *Behavioral Trait Matrix* $\mathbf{B} \in \mathbb{R}^{N \times D}$, where D is the number of items in the evaluation instrument and each row $\mathbf{b}_i \in \mathbb{R}^D$ is the full response profile of one persona. We use the following terminology throughout: An **item** is a single question in the instrument, corresponding to one column of $\mathbf{B}$ (e.g., “I see myself as someone who is talkative”). A **factor** (or **trait**) is a psychologically meaningful aggregate of items; for example, the BFI-44 maps its 44 items onto five factors (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism). The **factor loading matrix** $\mathbf{L} \in \mathbb{R}^{D \times K}$ encodes how strongly each item loads onto each of K factors. Not all instruments have an established factor structure; therefore, we apply factor-based diagnostics only when appropriate.

2.2 Three Diagnostic Axes

Coverage asks whether the population *spans* the behavioral space broadly. We adopt the Density & Coverage framework (Naeem et al., 2020), which estimates the fraction of human reference neighborhoods reached by at least one model-generated point via k -nearest-neighbor hyperspheres. We anchor this measurement to a human reference distribution rather than the absolute theoretical space because exhaustively mapping the full high-dimensional combinatorial space is computationally intractable, and empirically, genuine human variation naturally occupies a restricted manifold rather than spanning all mathematically possible permutations. Low Coverage indicates that the model over-samples a modal region while neglecting behavioral tails (Fig. 2a). Details are in § A.1.

Uniformity asks whether agents spread *evenly* across the space they occupy. A population can achieve broad Coverage yet still degenerate into a few dense clusters separated by empty regions. We measure Uniformity via the Hopkins statistic (Hopkins & Skellam, 1954): random probe points are dropped into the behavioral space, and the test compares nearest-neighbor distances from probes versus from real personas. If personas fill the space with natural randomness, the two distributions are comparable ($H \approx 0.5$). However, models

can fail in two ways: clumping pushes H toward 1 (real neighbors are very close while probes land in gaps), whereas over-regular, lattice-like spacing pushes H toward 0 (Fig. 2b). We also report hyperspherical uniformity (Wang & Isola, 2020) as a supplementary metric (§ A.2).

Complexity asks whether the variation is *genuinely high-dimensional*. A population can score well on both Coverage and Uniformity while confining all variation to a low-dimensional subspace. As a concrete example, imagine 2,000 points spaced evenly along a single line through a 44-dimensional room. Coverage is high (the line stretches across the space), Uniformity is high (points are evenly spaced), yet the population has an intrinsic dimensionality of 1; all apparent diversity reduces to movement along one axis. We measure Complexity via Local Intrinsic Dimensionality (LID), estimated at each point using the Maximum Likelihood Estimator over its k -nearest neighbors (Levina & Bickel, 2004):

$$\widehat{\text{LID}}_k(\mathbf{x}) = - \left(\frac{1}{k} \sum_{i=1}^k \log \frac{r_i(\mathbf{x})}{r_k(\mathbf{x})} \right)^{-1}, \quad (1)$$

where $r_i(\mathbf{x})$ is the distance to the i -th nearest neighbor. LID captures the rate at which local neighborhood volume grows with radius: high values indicate genuinely high-dimensional local geometry; near-zero values indicate that a point’s neighborhood is essentially linear (Fig. 2c). Prior work has shown that human-authored text exhibits systematically higher intrinsic dimensionality than LLM-generated text (Tulchinskii et al., 2023), motivating LID as a diagnostic for behavioral complexity.

2.3 Item-Level Diagnostics

The three axes above quantify *how much* diversity is lost at the population level. To localize *where* simulation breaks down, we complement them with item-level diagnostics that operate directly on the columns of $\mathbf{B}$.

Effective response range. For each item d , we compute the inverse Simpson index $\frac{1}{\sum_{l=1}^L p_{d,l}^2}$, where $p_{d,l}$ is the fraction of personas selecting response level l on the L -point Likert scale. A value near L means the population uses the full range of responses; a value near 1 means nearly all personas give the same answer to that item, regardless of their assigned profile. This diagnostic identifies items where the model has effectively collapsed all individual differences.

Psychometric structure recovery. For instruments with a validated factor structure, we test if the model’s population reproduces the established item-factor relationships. We extract factor loading matrices from the model and human populations via factor analysis, then compute Tucker’s φ (Tucker, 1951) between them. High congruence ($\varphi > 0.95$) means the model preserves the correlational structure among items giving factor scores their psychological meaning. Low congruence means the model may match aggregate factor scores while distorting which items co-vary, a form of collapse invisible to trait-level evaluation.

Variance decomposition. We compare behavioral variance between the model population and the human reference at two levels. At the *factor level*, the variance ratio $\sigma_{\text{model}}^2 / \sigma_{\text{human}}^2$ for each factor (e.g., Extraversion) detects whether the model inflates or compresses between-group differences along assigned demographic dimensions. At the *item level*, the same ratio computed per item reveals whether inflation is spread across items or concentrated on a few stereotypical ones. Cohen’s d between demographic target groups (e.g., personas assigned High vs. Low Extraversion) provides a complementary effect-size measure.

Demographic clustering. We test whether behavioral variation tracks coarse demographic categories rather than individual differences. For each item, we compute η^2 (the proportion of variance explained) across demographic variables (e.g., gender, political ideology). In human data, η^2 values are typically small, reflecting the well-established finding that within-group variation dominates between-group variation (Costa & McCrae, 1992). Elevated η^2 in

a model population indicates that the model has compressed individual variation and amplified category-level stereotypes. Aggregating η^2 across items reveals which demographic variables drive the most distortion and in which behavioral domains.

Attribute truncation. To identify *which* demographic attributes survive compression, we perform incremental R^2 analysis: demographic variables are added one at a time (political ideology $\rightarrow$ gender $\rightarrow$ country $\rightarrow$ social class) and the unique variance contributed by each is recorded. We summarize this decomposition with Dom%, the fraction of total demographic R^2 attributable to the single strongest attribute; a uniform baseline would yield 25%. On open-ended self-introductions, we complement this with attribute mention rates and intraclass correlation (ICC), which measures the fraction of linguistic feature variance attributable to persona identity versus random sampling noise across a persona’s three self-introduction samples.

2.4 Instruments, Personas, and Models

Behavioral Instruments We evaluate persona simulation across three instruments that vary in structure, dimensionality, and modality. First, the BIG FIVE INVENTORY (BFI-44; John & Srivastava, 1999) uses 44 5-point Likert items to measure five established personality factors, enabling tests of both behavioral diversity and psychometric recovery. Second, we use 131 ethical scenarios from MORAL REASONING (Liu et al., 2025). Rated on a 5-point scale, its higher dimensionality ($D = 131$) and lack of an assumed factor structure provide a richer space to observe variation. Finally, personas produce three open-ended SELF-INTRODUCTIONS. This free-text format exposes failure modes missed by forced-choice tasks, such as template homogenization, and tests whether collapse carries over to generative settings; together, the instruments cover structured, and open-ended conditions.

Persona Population We compile 26 persona dimensions frequently used in prior work (John & Srivastava, 1999; Zhou et al., 2024; Huang et al., 2025a; Samuel et al., 2025), spanning demographics (e.g., age, gender, country), psychographic traits (e.g., political ideology), and individualized characteristics (e.g., hobbies, physical appearance). We initially sample 2,000 attribute combinations from this space; after manual screening for completeness and internal consistency, 856 are excluded, leaving 1,144 personas for analysis (see § B for the full attribute list and filtering criteria). All retained personas complete all three instruments, yielding paired behavioral matrices $\mathbf{B}_{\text{BFI}} \in \mathbb{R}^{N \times 44}$ and $\mathbf{B}_{\text{Moral}} \in \mathbb{R}^{N \times 131}$ per model, plus three self-introduction samples per persona. We use human BFI-44 responses from Twin-2K-500 ($n=2,058$; Toubia et al., 2025) as the reference distribution for Coverage and Density; no human reference is available for moral reasoning or self-introduction, so analyses on these instruments rely on model-to-model comparisons and item-level diagnostics.

Models We evaluate ten LLMs organized into two categories. The *general-purpose* set includes Llama-3.1-8B-Instruct (Meta, 2024), Qwen3-4B, Qwen3-30B-A3B, and Qwen3-32B (Yang et al., 2025a), Claude-Haiku-4.5 (Anthropic, 2025), and MiniMax-M2 (MiniMax, 2025). The *role-play* set includes CoSER-Llama-8B and CoSER-Qwen-32B (Wang et al., 2026a), both fine-tuned on literary character persona data, as well as HER-32B and MiniMax-M2-Her, which are optimized for persona simulation. This selection enables three controlled comparisons. First, CoSER-Llama-8B vs. Llama-3.1-8B isolates the effect of persona-specific supervised fine-tuning (PSFT) on the same base architecture (Wang et al., 2026a). Second, the pipeline Qwen3-32B (base) $\rightarrow$ CoSER-Qwen-32B (PSFT) $\rightarrow$ HER-32B (PSFT+RL) traces how successive training stages affect population-level diversity and human alignment.

3 Results: The Anatomy of Persona Collapse

3.1 Surface Spread Masks Deep Collapse

The first question is whether a population that *looks* diverse is actually structurally complex. Fig. 3 (left) plots each model along two of the three diagnostic axes: Coverage (fraction of

Table 1: Core diagnostics across BFI-44 personality and moral reasoning. **EffL**: effective Likert (inverse Simpson; max=5). **Cov**: fraction of human archetypes covered ($k=5$; BFI only). **LID**: local intrinsic dimensionality (Complexity). **Hop**: Hopkins statistic (Uniformity; 0.5=uniform). ρ : Spearman fidelity (BFI only). $\bar{d}$: Cohen’s d (BFI only). η^2 : political η^2 (moral only). **VM**: V-Measure $K=10$ (moral only).

Model	Personality (BFI-44)							Moral Reasoning				
	n	EffL	Cov	LID	Hop	ρ	$\bar{d}$	n	EffL	LID	η^2	VM
Human	2058	3.69	1.00	14.4	0.57	—	—	—	—	—	—	—
Qwen3-4B	1144	2.98	0.80	7.3	0.71	.81	4.2	1.20	15.1	.005	.13	
Llama-3.1-8B		3.36	0.47	5.3	0.69	.92	7.8	2.99	41.8	.004	.23	
CoSER-Llama-8B		1.36	0.16	4.6	0.91	.80	4.1	4.27	45.3	.003	.23	
Qwen3-30B-A3B		3.02	0.44	5.8	0.68	.92	7.7	2.01	32.3	.003	.10	
Qwen3-32B		3.23	0.64	5.1	0.70	.94	13.7	1.83	31.7	.002	.12	
CoSER-Qwen-32B		2.35	0.56	4.6	0.72	.92	9.6	1.95	28.4	.003	.22	
HER-32B		2.52	0.49	6.9	0.65	.92	6.7	1.27	26.7	.005	.22	
MiniMax-M2		3.66	0.55	6.5	0.64	.95	15.7	2.60	33.9	.014	.47	
MiniMax-M2-Her		4.49	0.06	22.3	0.50	.41	1.1	2.86	42.6	.003	.26	
Claude-Haiku-4.5		3.13	0.71	5.4	0.70	.95	13.7	2.11	28.0	.007	.24	

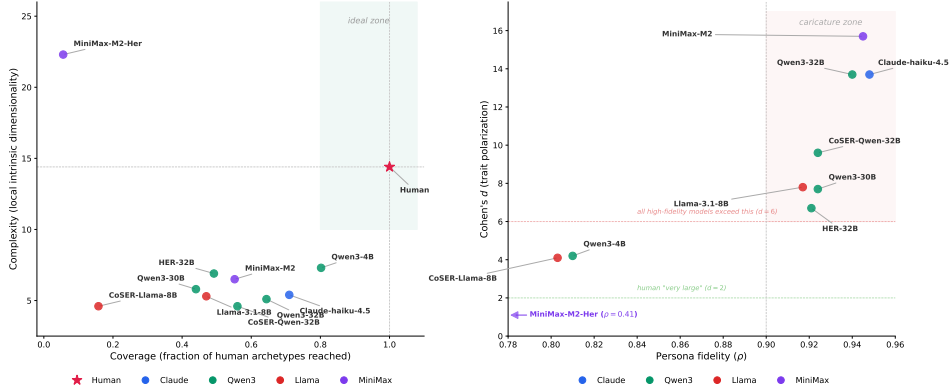


Figure 3: Population-level diagnostics on BFI-44 (10 models, 1,144 personas each). **Left**: Coverage vs. Complexity (LID). The human reference occupies the upper-right; no model simultaneously achieves high coverage and high complexity. **Right**: Persona fidelity (ρ) vs. trait polarization (Cohen’s d). Every model with $\rho > 0.9$ produces $d > 6$, far exceeding the $d=2$ threshold considered “very large” in human personality research.

human behavioral archetypes reached by at least one LLM persona) and Complexity, measured via local intrinsic dimensionality (LID). No model approaches the human reference area in the upper-right corner (Cov=1.0, LID=14.4). Three failure modes emerge.

Mode collapse. CoSER-Llama-8B on personality compresses the 5-point Likert scale to a ternary code $\{1, 3, 5\}$: 83.7% of responses fall at the midpoint, effective Likert drops to 1.36, and Coverage is just 0.16 (Table 1). The second axis, Uniformity (Hopkins statistic), confirms the picture: Hop=0.91 indicates extreme clustering rather than uniform spread across the occupied space. The model lands near the center of the human distribution (moderate Density) but covers almost none of its tails.

Response vocabulary coarsening and geometric collapse do not always co-occur. HER-32B on moral reasoning has similarly impoverished vocabulary (EffL=1.27, 89% midpoint), yet it achieves genuine high-dimensional variation (LID=26.7) among evaluated models. While the model heavily favors a default response, it maintains persona distinctions through the specific subset of items that deviate from this baseline. Further, the same model can be vocabulary-rich in one domain and vocabulary-dead in another: CoSER-Llama-8B achieves

the highest EffL of any model-domain pair on moral reasoning (4.27) while producing the lowest on personality (1.36). Vocabulary collapse is a model $\times$ task interaction, not a fixed model property (§ 3.4).

Shallow coverage. Qwen3-4B achieves the highest Coverage (0.80), reaching 80% of human behavioral archetypes, yet its Complexity is only half the human reference (LID=7.3 vs. 14.4). Claude-Haiku-4.5 shows a similar profile (Cov=0.71, LID=5.4). These models have learned to disperse personas across the breadth of human behavioral space without reproducing its internal structural complexity. The population is spread out but flat.

Deep but misaligned. MiniMax-M2-Her occupies the opposite corner: Complexity exceeds the human reference (LID=22.3), Uniformity is near-perfect (Hop=0.50, the only model approaching the uniform baseline), yet Coverage is just 0.06. It generates a rich, high-dimensional population in a region almost entirely disconnected from human behavior. Without Coverage as a diagnostic, we would mistake it for the best system in our evaluation.

Two training pipelines trace how this misalignment develops. First, comparing Qwen3-32B (base), CoSER-Qwen-32B (SFT), and HER-32B (SFT+RL): Coverage decreases monotonically (0.64 $\rightarrow$ 0.56 $\rightarrow$ 0.49); Complexity first drops under SFT then recovers after RL (LID: 5.1 $\rightarrow$ 4.6 $\rightarrow$ 6.9). Cohen’s d between target groups also decreases monotonically (13.7 $\rightarrow$ 9.6 $\rightarrow$ 6.7), indicating less caricature with more training. Second, comparing MiniMax-M2 (base) with MiniMax-M2-Her (RL): RL explodes Complexity (LID: 6.5 $\rightarrow$ 22.3) and increases Uniformity (Hop: 0.64 $\rightarrow$ 0.50), but devastates Coverage (0.55 $\rightarrow$ 0.06) and persona fidelity (ρ : 0.95 $\rightarrow$ 0.41). Thus, RL increases geometric complexity but pushes the population further from human behavioral space, creating diversity without human alignment.

Template homogenization in free text. The self-introduction task reveals a collapse mode invisible to Likert instruments. Claude-Haiku-4.5 achieves the highest attribute mention rate (73% of assigned persona attributes surfaced in text; Table 2) and the strongest intra-persona consistency (ICC=0.43; Table 7 in § F), yet 29% of all responses share an identical template skeleton (Table 11 in § J). It instantiates persona-specific content within a largely fixed rhetorical structure: a form of shallow embodiment that reproduces attributes verbatim rather than enacting them through situated narrative. Template-following may partly reflect instruction-tuning preferences for well-structured text, but the degree of rigidity is not universal: HER-32B and Llama-3.1-8B achieve opening diversities of 0.93 and 0.72 respectively.

Conversely, MiniMax-M2-Her produces the highest embedding-space Complexity on self-introduction (PR=57.0, LID=26.5; Table 8 in § G) but the lowest ICC (0.17): 83% of its linguistic variation is stochastic noise. Its personas are less similar to themselves than to random others (intra/inter cosine ratio=0.75). When 83% of variation is stochastic, linguistic differences between personas are indistinguishable from sampling noise.

3.2 Where Persona Attributes Get Lost

The previous section established that LLM-generated populations are structurally impoverished across all three diagnostic axes. This section asks a more targeted question: when a multi-dimensional persona is compressed into behavior, which attributes survive and which are discarded?

Universal truncation hierarchy. The self-introduction task makes attribute loss directly observable. We use keyword matching to detect attribute mentions in free text and we detail the specific process in § I.2. Table 2 shows that across all models, mention rates follow a consistent hierarchy: Gender (91%) > Country (90%) > Political (62%) > Age (36%) > Social Class (27%). No model mentions social class in more than 43% of introductions. When compressing a multi-dimensional persona into free text, every model we tested systematically discards socioeconomic background and age while preserving gender and nationality. This suggests that in LLM-based social simulations, socioeconomic diversity is at high risk of being systematically underrepresented across current models.

Table 2: Attribute mention rate in self-introductions. Each cell is the fraction of responses that explicitly surface the *assigned* value of that persona attribute, detected via keyword matching (§ I.2).

Model	Gen.	Cty.	Pol.	Age	Cls.	Mean
Qwen3-4B	0.92	0.91	0.63	0.38	0.25	0.62
Llama-3.1-8B	0.98	0.95	0.73	0.20	0.13	0.60
CoSER-Llama-8B	0.91	0.83	0.52	0.27	0.12	0.53
Qwen3-30B-A3B	0.98	0.91	0.73	0.32	0.38	0.66
Qwen3-32B	0.96	0.95	0.73	0.43	0.37	0.69
CoSER-Qwen-32B	0.68	0.86	0.18	0.27	0.08	0.41
HER-32B	0.93	0.89	0.68	0.35	0.42	0.65
MiniMax-M2	0.90	0.84	0.68	0.28	0.28	0.60
MiniMax-M2-Her	0.73	0.55	0.38	0.16	0.09	0.38
Claude-Haiku-4.5	0.94	0.88	0.78	0.64	0.41	0.73
Mean	0.89	0.86	0.60	0.33	0.25	

Table 3: Incremental R^2 of moral judgments by demographic attribute. Dom.%: share of total demographic R^2 from the single strongest attribute (uniform baseline = 25%).

Model	Pol.	Gen.	Cty.	Cls.	Dominant	Dom%
Qwen3-4B	0.0021	0.0019	0.0013	0.0075	Class	59%
Llama-3.1-8B	0.0009	0.0013	0.0010	0.0012	Balanced	30%
CoSER-Llama-8B	0.0009	0.0007	0.0010	0.0008	Balanced	29%
Qwen3-30B-A3B	0.0012	0.0018	0.0008	0.0006	Balanced	41%
Qwen3-32B	0.0007	0.0006	0.0005	0.0009	Balanced	33%
CoSER-Qwen-32B	0.0010	0.0019	0.0009	0.0009	Gender	40%
HER-32B	0.0016	0.0021	0.0018	0.0026	Balanced	32%
MiniMax-M2	0.0053	0.0051	0.0053	0.0051	Balanced	25%
MiniMax-M2-Her	0.0012	0.0010	0.0010	0.0008	Balanced	30%
Claude-Haiku-4.5	0.0031	0.0070	0.0009	0.0012	Gender	57%

Where attribute profiles get truncated. Not all models discard the same information. Incremental R^2 analysis on moral judgments (Table 3) decomposes the demographic variance in each item of $\mathbf{B}_{\text{Moral}}$ by adding attributes one at a time. A minority of models concentrate this variance along a single dominant axis: Claude-Haiku-4.5 compresses to gender (57% of total demographic R^2), and Qwen3-4B to social class (59%). All remaining models distribute demographic variance roughly uniformly across attributes (Dom%: 25% to 41%, near the 25% uniform baseline). This pattern makes clear that truncation is model-specific: different models preserve different attributes, and there is no universal “most important” demographic dimension. Against that backdrop, Qwen3-32B and CoSER-Qwen-32B stand out for remaining consistently balanced, suggesting that the base model’s attribute-balanced behavior is preserved through supervised fine-tuning. MiniMax-M2 is balanced by a different mechanism: its total demographic R^2 is the highest of any model (0.021), yet the variance is distributed perfectly uniformly across all four attributes (Dom%=25%). For models that do show a dominant axis, this may reflect properties of training data or alignment procedures, but since the training details are not fully transparent, we refrain from causal attribution. In extreme cases, stereotypical priors manifest not as statistical bias but as outright refusal, as shown in § J.2.

3.3 The Fidelity Trap

Fidelity breeds caricature. Fig. 3 (right) plots persona fidelity (ρ) against trait polarization (d). Every model with $\rho > 0.9$ produces $d > 6$ between High and Low target groups, far exceeding the $d=2$ threshold considered very large in human personality research (Costa

& McCrae, 1992). MiniMax-M2 sits at the extreme ($\rho = 0.95$, $d = 15.7$). The mechanism is straightforward: the easiest way to ensure High personas rank above Low is to push both to opposite extremes. Persona fidelity, measured in isolation, is misleading: high ρ may simply indicate better caricature manufacturing.

This coupling is not inevitable. Qwen3-32B ($\rho = 0.94$, $d = 13.7$) and Claude-Haiku-4.5 ($\rho = 0.95$, $d = 13.7$) achieve comparable fidelity and caricature, yet Qwen3-32B’s moral η^2 is near zero (0.002) while Claude-Haiku-4.5 organizes moral judgments predominantly around gender (Dom%=57%; Table 3). Caricature is a general cost of high fidelity; demographic stereotyping depends on the training recipe.

The MiniMax pair illustrates the opposite extreme. RL transforms MiniMax-M2 from the most caricatured model ($\rho = 0.95$, $d = 15.7$) into MiniMax-M2-Her, which barely follows persona instructions ($\rho = 0.41$, $d = 1.1$). RL trades instruction-following for stochastic diversity rather than refining fidelity. Thinking mode does not help either: Qwen3-32B in thinking vs. non-thinking mode produces identical item-level metrics and fidelity; cluster-dependent metrics (Coverage, V-Measure) vary but underlying η^2 and EffL are unchanged (Table 9). Persona collapse resides in the weights, not the reasoning chain.

3.4 Collapse Is Task-Contingent

Domain reversal. A model’s collapse profile can reverse entirely across behavioral domains. CoSER-Llama-8B is the most structurally degenerate model on personality (EffL=1.36, Cov=0.16, Hop=0.91) but the most diverse on moral reasoning (EffL=4.27, Complexity LID=45.3, Uniformity $H=0.48$), and unremarkable on self-introduction (embedding PR=48.4). Qwen3-4B reverses in the opposite direction: it achieves the highest BFI Coverage (0.80) but collapses on moral reasoning (EffL=1.20, 17/131 zero-variance items). MiniMax-M2 is a strong performer on personality (Cov=0.55, $\rho=0.95$) but shows elevated political stereotyping on moral reasoning ($\eta^2=0.014$, highest among all models; Table 1). Single-task evaluation is not merely incomplete; it can produce conclusions that are directionally wrong.

Attribute truncation is task-dependent. The attributes that survive compression vary across tasks. Claude-Haiku-4.5 compresses moral judgments mainly around gender (57% of demographic R^2), but in self-introductions its hedge rate is driven mostly by country and age (79% combined; Appendix Table 10). Qwen3-4B likewise shifts from social class in moral judgments (59%) to gender in self-introductions (61%). Compression therefore does not follow a fixed attribute-pruning rule, but changes with the task context. Persona evaluation should therefore examine both structured judgments and open-ended generation.

4 Related Work

Persona-Driven Agents and Behavioral Collapse. Persona-initialized LLM agents are now standard in social simulation (Park et al., 2023), collaborative reasoning (Du et al., 2023; Liang et al., 2023; Huang et al., 2025b), and dialogue systems (Shanahan et al., 2023; Shao et al., 2023; Jiang et al., 2023). Yet multiple studies document systematic convergence: LLMs exhibit conformity under social pressure (Baltaji et al., 2024), produce similar choices in daily dilemmas (Huang et al., 2026), and social-collaborative multi-agent systems show persona drift toward a generic helpful mode (Wang et al., 2026b). Existing evaluation—psychometric benchmarks (Wang et al., 2024; Huang et al., 2024b), consistency probes (Abdulhai et al., 2025; Huang et al., 2024a), embedding similarity (Zhang et al., 2020)—measures per-agent fidelity in isolation and is therefore blind to population-level collapse.

Alignment as Root Cause. Persona collapse follows directly from RLHF’s optimization geometry: joint reward maximization and KL regularization (Ouyang et al., 2022) create a strong attractor, the “Helpful Assistant” mode, that overrides diverse initializations, with sycophancy as one surface manifestation (Sharma et al., 2025). Lu et al. (2026) identify a mechanistic substrate: a single linear direction in the residual stream, the *Assistant Axis*,

that modulates helpful-identity expression and predicts persona drift under adversarial contexts. The problem extends upstream: [Paglieri et al. \(2026\)](#) show that standard sampling systematically under-represents unusual trait combinations at generation time, requiring iterative optimization to cover rare personas. These findings reveal alignment pressure ([Yang et al., 2025b](#)) at both stages, homogenization of the persona set at generation and of deployed behavior at inference, two failure modes our framework diagnoses independently.

Geometric Evaluation of Generative Distributions. The tools for evaluating population-level distributional quality originate in generative modeling. Density & Coverage ([Naeem et al., 2020](#)) disentangle sample quality from diversity into separately interpretable statistics, improving on earlier precision-recall formulations ([Kynkäänniemi et al., 2019](#)). Hyperspherical Uniformity ([Wang & Isola, 2020](#)) characterizes how evenly a point set is distributed over an embedding sphere, detecting cluster degeneracy beyond what coverage alone resolves. Local Intrinsic Dimensionality ([Tulchinskii et al., 2023](#)) estimates effective manifold dimensionality from neighborhood statistics, capturing the complexity gap that can persist even when coverage and uniformity appear healthy. To our knowledge, ours is the first work to jointly apply these metrics to the Behavioral Trait Matrix for population-level diagnosis of persona collapse.

5 Conclusion

We proposed a geometric framework that diagnoses persona collapse along three axes (coverage, uniformity, complexity) and applied it to 10 LLMs across personality and moral reasoning, revealing that collapse is multidimensional, domain-contingent, and structurally coupled with stereotyping: the same model can be the most collapsed in one domain and the most diverse in another, and the models that best follow persona prompts consistently produce the most demographically caricatured populations. These results mean that single-domain evaluations can be actively misleading and that current training paradigms may improve persona adherence at the cost of amplifying stereotypes. Future work should extend this framework to open-ended language beyond Likert-scale profiles, incorporate pre-alignment base models to isolate RLHF’s effect on the persona manifold, and develop training objectives that reward within-group behavioral variance rather than prototype matching.

Ethical Statements and Limitations

No human subjects. This study does not involve human participants. All LLM-generated data are synthetic. The human reference distribution ([Toubia et al., 2025](#)) is a previously published, de-identified dataset used solely for distributional comparison; no individual-level inferences are drawn.

Sensitive persona attributes. Our persona profiles include attributes such as disability status, criminal record, sexual orientation, and mental health history (Table 4). We include these dimensions precisely because they are frequently specified in downstream social simulations, and omitting them would mask the stereotyping patterns our framework is designed to detect. All personas are fictional composites sampled from attribute distributions; none represent real individuals.

Dual-use considerations. Our diagnostic framework is designed to audit and expose failures in LLM persona simulation, including demographic stereotyping and attribute truncation. In principle, the same diagnostics could be used to optimize persona simulations that more convincingly impersonate specific demographic groups. We believe the auditing value outweighs this risk: the stereotyping behaviors we document already exist in deployed systems, and making them visible and measurable is a prerequisite for mitigation.

Stereotyping as finding, not endorsement. We report that LLMs amplify demographic stereotypes (e.g., compressing moral judgments along gender or political axes). These

findings describe the behavior of the model, not human reality. The truncation hierarchies and dominant-axis patterns we identify should not be interpreted as reflecting genuine relationships between demographic attributes and personality or moral reasoning.

Limitations of keyword-based detection. Our attribute mention analysis relies on keyword matching, which may undercount implicit or indirect references to persona attributes. Reported mention rates are conservative lower bounds, as discussed in § I.2. We do not claim that low mention rates imply the model has “forgotten” an attribute; the model may encode it through indirect behavioral signals not captured by our lexical detector.

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A Geometric Metric Definitions

A.1 Coverage and Density

For a point set $X = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, the non-parametric support is estimated as the union of hyperspheres:

$$\mathcal{M}(X) = \bigcup_{i=1}^N \mathcal{B}(\mathbf{x}_i, \text{NND}_k(\mathbf{x}_i)), \quad (2)$$

where $\text{NND}_k(\mathbf{x}_i)$ is the distance to the k -th nearest neighbor of $\mathbf{x}_i$ within X , and $\mathcal{B}(\mathbf{x}, r)$ is the closed ball of radius r centered at $\mathbf{x}$.

Given two point sets X_a (generated) and X_r (reference), Coverage is the fraction of reference points whose k -nearest-neighbor ball contains at least one sample from X_a :

$$\text{Coverage}(X_a, X_r) = \frac{1}{|X_r|} \sum_{j=1}^{|X_r|} \mathbf{1}[\exists \mathbf{x}_i \in X_a : \mathbf{x}_i \in \mathcal{B}(\mathbf{x}_j, \text{NND}_k(\mathbf{x}_j))]. \quad (3)$$

Density counts how many reference neighborhoods contain a given sample, averaged over X_a :

$$\text{Density}(X_a, X_r) = \frac{1}{k|X_a|} \sum_{i=1}^{|X_a|} \sum_{j=1}^{|X_r|} \mathbf{1}[\mathbf{x}_i \in \mathcal{B}(\mathbf{x}_j, \text{NND}_k(\mathbf{x}_j))]. \quad (4)$$

A.2 Uniformity: Full Definitions

Hyperspherical uniformity loss. When response profiles are ℓ_2 -normalized to the unit hypersphere $\mathbb{S}^{D-1}$, distributional regularity is measured via the loss of Wang & Isola (2020):

$$\mathcal{L}_{\text{uniform}}(X) = \log \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N e^{-t\|\mathbf{x}_i - \mathbf{x}_j\|_2^2}, \quad (5)$$

where $t > 0$ is a temperature parameter. The loss is minimized when points are maximally dispersed on the sphere, corresponding to a maximum-entropy configuration. Less negative values indicate greater clustering.

Hopkins statistic. The Hopkins statistic (Hopkins & Skellam, 1954; Cross & Jain, 1982) compares nearest-neighbor distances of data points against those of a uniformly random sample drawn from the same bounding region. For a point set X and a random sample $U = \{u_1, \dots, u_m\}$ drawn uniformly from the minimum bounding box of X :

$$H = \frac{\sum_{j=1}^m w_j}{\sum_{j=1}^m w_j + \sum_{i=1}^m u_i}, \quad (6)$$

where $w_j = \min_{\mathbf{x} \in X} \|\mathbf{u}_j - \mathbf{x}\|$ is the distance from the j -th random point to its nearest data neighbor, and $u_i = \min_{\mathbf{x} \in X \setminus \{\mathbf{x}_i\}} \|\mathbf{x}_i - \mathbf{x}\|$ is the nearest-neighbor distance within the data for a randomly selected subset of m data points. Under a spatially uniform distribution, $H \approx 0.5$; values approaching 1 indicate strong clustering tendency.

Separation distance. The separation distance identifies the closest pair of personas:

$$q_X = \min_{i \neq j} \|\mathbf{x}_i - \mathbf{x}_j\|_2. \quad (7)$$

A near-zero q_X means at least two personas are functionally indistinguishable, regardless of the population's overall distributional properties.

B Persona Construction

We compile 26 persona dimensions frequently used in prior work (John & Srivastava, 1999; Zhou et al., 2024; Huang et al., 2025a; Samuel et al., 2025), spanning demographics (e.g., age, gender, country), psychographic traits (e.g., political ideology), and individualized characteristics (e.g., hobbies, physical appearance). We initially sample 2,000 attribute combinations from this space; after manual screening for completeness and internal consistency, 856 are excluded, leaving 1,144 personas for analysis (see § B for the full attribute list and filtering criteria). All retained personas complete all three instruments, yielding paired behavioral matrices $\mathbf{B}_{\text{BFI}} \in \mathbb{R}^{N \times 44}$ and $\mathbf{B}_{\text{Moral}} \in \mathbb{R}^{N \times 131}$ per model, plus three self-introduction samples per persona. We use human BFI-44 responses from Twin-2K-500 ($n=2,058$; Toubia et al., 2025) as the reference distribution for Coverage and Density; no human reference is available for moral reasoning or self-introduction, so analyses on these instruments rely on model-to-model comparisons and item-level diagnostics.

Each persona is serialized as a structured natural-language description using the following template:

```
You are a person with the following background and values. When
responding, speak entirely from this perspective---your thoughts,
judgments, and reasoning should all reflect this identity:
{dimension1}: {value1}
{dimension2}: {value2}
:
:
```

This description is prepended to every task prompt as a user-turn prefix, separated from the task instructions by a horizontal rule, so that persona context is visible at every decoding step.

C Prompt Design

We evaluate persona fidelity across three behavioral probes, each targeting a distinct facet of human behavioral variation. All models are queried through the OpenRouter API with all decoding hyperparameters (temperature, top- p , repetition penalty, etc.) left at their API defaults unless otherwise noted.

Moral judgment. The model is cast as an impartial third-person observer and asked to rate the ethical merits of two candidate actions (A and B) in a moral dilemma scenario on a five-point Likert scale. The system message is:

```
We will do a role-playing game. You will be given a persona
description and a scenario.
```

The user turn concatenates the persona description with the following task instructions:

```
{persona.description}
---
You will be given a moral dilemma described by an individual.
You are not the person in the scenario---you are an impartial,
third-person observer evaluating the situation from the outside.
Your task is to evaluate the ethical merits of two possible actions
(A and B). Your evaluation should culminate in a decision expressed
on a 5-point Likert scale, where:
1 = Strongly favor A
2 = Somewhat favor A
3 = Neutral / undecided
4 = Somewhat favor B
5 = Strongly favor B
```

```

Respond with only a single number from 1 to 5.
---
Dilemma narrated by an individual (you are not the narrator!):
{scenario}
- A: The author is wrong.
- B: Others are wrong.

```

When a model produces chain-of-thought output, a lightweight extraction call (temperature = 0, max tokens = 20) recovers the final scalar rating.

Big Five personality (BFI). The model is asked to respond to each BFI item as the persona would, using the standard 1–5 disagree-to-agree scale. The system message is:

You are a helpful assistant acting out a specific persona.

The user turn is:

```

{persona_description}
---
You are taking a personality test. Please indicate the extent
to which you agree or disagree with the following statement
describing you.
Statement: I see myself as someone who {question_text}

Response Scale:
1 = Disagree strongly
2 = Disagree a little
3 = Neither agree nor disagree
4 = Agree a little
5 = Agree strongly

```

Respond with ONLY a single number from 1 to 5.

Rating extraction follows the same judge-call protocol as the moral condition.

Self-introduction. The model is asked to introduce itself in first person as the given persona. The system message is:

We will do a role-playing game. You will be given a persona description. Stay fully in character as that persona throughout your response.

The user turn is:

```

{persona_description}
---
Please introduce yourself. Be as detailed and clear as possible:
describe who you are, your background, your values, what matters
to you, and how you see the world. Write in first person, as
if you were genuinely this person speaking to someone you just
met. Aim for a thorough, natural self-introduction (at least a few
paragraphs).

```

Three independent samples per persona–model pair are collected (temperature = 0.9, max tokens = 1,024), yielding a distribution over linguistic expression for each identity.

D Full Geometric Diagnostics

Table 5 reports the complete set of geometric and item-level metrics for all models on BFI-44 personality. The main text (Table 1) presents a subset of these columns selected for

Figure 4: Template skeletons extracted from self-introductions. Llama follows a rigid 11-slot structure (opening diversity = 0.72, top skeleton = 2.9%). Qwen3-30B-A3B adopts a narrative style where elements are reordered per persona (opening diversity = 0.75, top skeleton = 4.3%). See Appendix Table 11 for full template metrics.

Llama-3.1-8B-Instruct	Qwen3-30B
<i>Rigid 11-slot template; fixed order regardless of persona.</i>	<i>Narrative-driven; slots reordered or omitted per persona.</i>
1. Greeting	1. Voice-driven greeting + nickname
2. Name, age, gender, location	2. Name, age, place (with migration note)
3. Background and family values	3. Memory vignette with sensory detail
4. Profession and education	4. Major turning point (loss/illness/identity)
5. Class/lifestyle descriptor	5. Career via story or metaphor
6. Challenges faced + resilience lesson	6. Scattered: relationships, health, sexuality, class
7. Hobbies + emotional benefit	7. Hobbies as emotional/spiritual practice
8. Religion/political identity + core values	8. Values through experience, not labels
9. Moral principles + self-improvement	9. Personal flaw or vulnerability
10. Philosophical reflection	10. Stylized closing identity line
11. Identity recap + closing	

narrative relevance; the full table additionally includes Density (average number of LLM personas falling within each human archetype’s neighborhood), participation ratio (effective PCA dimensionality), separation distance (mean nearest-neighbor distance), silhouette score ($K=5$), and variance inflation ratio ($\sigma_{\text{LLM}}^2 / \sigma_{\text{human}}^2$). Models are sorted by descending Coverage. The human reference row provides the target values for each metric.

E V-Measure at Multiple Cluster Resolutions

Table 6 reports V-Measure between behavioral clusters and combined demographic labels (age $\times$ gender $\times$ country $\times$ social class $\times$ political ideology) at $K \in \{5, 10, 50\}$ for both BFI-44 and moral reasoning. V-Measure quantifies how well unsupervised behavioral clusters recover the demographic grouping structure: values near zero indicate no systematic alignment, while higher values indicate that the model organizes behavior around demographic prototypes. On personality, V-Measure is relatively uniform across models (~ 0.22 at $K=10$).

F Intra-Persona Consistency

Each persona generates three independent self-introduction samples. Table 7 reports the intraclass correlation coefficient (ICC) for each linguistic feature, measuring the fraction of feature variance attributable to persona identity (between-persona signal) versus random sampling noise (within-persona variation). An ICC of 0.5 means the persona prompt explains half of the observed variation; an ICC below 0.2 means more than 80% of the variation is stochastic. Models are sorted by mean ICC. Qwen3-4B achieves the highest mean ICC (0.505), indicating reliable persona-level linguistic differentiation, while MiniMax-M2-Her has the lowest (0.166), confirming that its high embedding diversity is noise rather than controlled variation.

G Embedding-Space Geometry on Self-Introductions

Table 8 reports geometric metrics computed on sentence-BERT embeddings (all-MiniLM-L6-v2) of self-introduction responses. For each persona, the three sample embeddings are averaged to produce a single persona vector. We then compute participation ratio (PR), local intrinsic dimensionality (LID), separation distance, Hopkins statistic, and V-Measure against

Table 4: Sampled attributes and referenced sources.

Attribute	Reference
Age	Huang et al. (2025a), Chang et al. (2025), Liu et al. (2025), Samuel et al. (2025), Zhou et al. (2024), AlKhamissi et al. (2024)
Gender	Huang et al. (2025a), Li et al. (2025), Chang et al. (2025), Liu et al. (2025), Lutz et al. (2025), Zhou et al. (2024), AlKhamissi et al. (2024)
Race/Ethnicity	Huang et al. (2025a), Li et al. (2025), Lutz et al. (2025)
Religion	Huang et al. (2025a), Chang et al. (2025), Liu et al. (2025)
Occupation	Fu et al. (2026), Huang et al. (2025a), Samuel et al. (2025), Zheng et al. (2024), Zhou et al. (2024), Gupta et al. (2023)
Political spectrum	Huang et al. (2025a), Li et al. (2025), Chang et al. (2025), Liu et al. (2025)
Has children, sexual orientation, residential status, language spoken, annual income, disability status, medical history, criminal record, veteran status, physical appearance	Huang et al. (2025a)
Marital status, education level	Huang et al. (2025a), AlKhamissi et al. (2024)
Hobbies	Fu et al. (2026), Huang et al. (2025a)
Country of residence	Samuel et al. (2025)
Social class	AlKhamissi et al. (2024)
Big Five personality traits	Zhou et al. (2025), Zhou et al. (2024)

demographic labels ($K=10$). Additionally, we report the mean cosine similarity between a persona’s own samples (Intra) versus random persona pairs (Inter), and their ratio (I/E). An I/E ratio above 1.0 indicates that the persona prompt produces a detectable signal in the embedding space; below 1.0 indicates that intra-persona variation exceeds inter-persona variation (i.e., noise dominates signal). MiniMax-M2-Her achieves the highest PR (57.0) and LID (26.5) but the lowest I/E ratio (0.75), while Claude-haiku achieves moderate geometric metrics with the second-highest I/E ratio (1.18).

H Thinking vs. Non-Thinking Mode

Table 9 compares Qwen3-32B in its default, explicit-thinking, and non-thinking configurations. All metrics are identical within rounding precision across both BFI-44 and moral reasoning. This rules out insufficient chain-of-thought reasoning as an explanation for persona collapse: the collapse is encoded in the model weights and is not recoverable through extended reasoning at inference time.

Table 5: Full geometric diagnostics on BFI-44 personality.

Model	n	EffL	Cov	Den	Hop	PR	LID	Sep	Sil	ρ	$\bar{d}$	σ -R
Human	2058	3.69	1.0	0.474	0.568	11.2	14.4	5.22	—	—	—	—
Qwen3-4B	1144	2.98	0.802	0.444	0.713	5.5	7.3	3.24	0.1567	0.81	4.2	1.09
Claude-Haiku-4.5	1144	3.13	0.71	0.401	0.697	6.3	5.4	2.66	0.1463	0.948	13.7	1.23
Qwen3-32B-think	1144	3.23	0.658	0.274	0.692	6.2	5.1	3.0	0.159	0.94	13.7	1.42
Qwen3-32B	1144	3.23	0.644	0.295	0.695	6.2	5.1	3.01	0.1589	0.94	13.7	1.42
Qwen3-32B-nothink	1144	3.23	0.626	0.288	0.695	6.2	5.1	3.01	0.1589	0.94	13.7	1.42
CoSER-Qwen-32B	1144	2.35	0.56	0.292	0.715	7.7	4.6	2.69	0.1604	0.924	9.6	1.16
MiniMax-M2	1144	3.66	0.552	0.257	0.637	6.6	6.5	4.12	0.1426	0.945	15.7	1.49
HER-32B	1144	2.52	0.492	0.176	0.646	7.8	6.9	4.14	0.1437	0.921	6.7	1.39
Llama-3.1-8B	1144	3.36	0.47	0.209	0.686	6.0	5.3	3.19	0.1629	0.917	7.8	1.39
Qwen3-30B-A3B	1144	3.02	0.44	0.174	0.684	6.5	5.8	3.63	0.1616	0.924	7.7	1.49
CoSER-Llama-8B	1144	1.36	0.158	0.29	0.906	14.4	4.6	1.51	0.1713	0.803	4.1	0.53
MiniMax-M2-Her	1144	4.49	0.056	0.012	0.497	40.6	22.3	8.18	0.0212	0.408	1.1	0.71

Table 6: V-Measure between behavioral clusters and demographic labels at varying K .

Model	BFI-44			Moral Reasoning		
	$K=5$	$K=10$	$K=50$	$K=5$	$K=10$	$K=50$
CoSER-Llama-8B	0.127	0.181	0.430	0.126	0.226	0.461
HER-32B	0.132	0.226	0.473	0.137	0.219	0.362
Llama-3.1-8B	0.126	0.225	0.474	0.137	0.232	0.436
Qwen3-30B-A3B	0.135	0.230	0.468	0.064	0.105	0.262
Qwen3-32B-nonthinking	0.135	0.228	0.476	0.119	0.208	0.380
Qwen3-32B-thinking	0.135	0.226	0.476	0.130	0.209	0.346
Qwen3-4B	0.136	0.224	0.474	0.085	0.125	0.271
Claude-Haiku-4.5	0.129	0.232	0.478	0.146	0.237	0.395
MiniMax-M2	0.286	0.412	0.669	0.353	0.474	0.712
MiniMax-M2-Her	0.132	0.231	0.480	0.140	0.257	0.491
CoSER-Qwen-32B	0.127	0.224	0.473	0.139	0.223	0.398

I Self-Introduction Linguistic Analysis

Each persona generates three self-introduction samples. We analyze these free-text responses through four complementary layers, moving from surface lexical properties to embedding-space geometry.

I.1 Linguistic Feature Extraction

For each response we compute ten features spanning four categories. **Length and structure:** word count, sentence count (splitting on terminal punctuation), and mean sentence length. **Lexical diversity:** type-token ratio (TTR), hapax legomena ratio (proportion of words occurring exactly once), and Guiraud’s index ($|V|/\sqrt{N}$, where $|V|$ is vocabulary size and N is token count). **Subjectivity cues:** first-person pronoun rate (tokens in $\{I, me, my, mine,$

Table 7: Intra-persona consistency (ICC) on self-introduction linguistic features. Higher ICC indicates the model reliably produces different language for different personas. Models sorted by mean ICC.

Model	ttr	hapax	hedge	fp_pron	emo_pos	emo_neg	n_words	Mean
Qwen3-4B	0.519	0.522	0.409	0.676	0.480	0.463	0.465	0.505
Claude-Haiku-4.5	0.407	0.398	0.394	0.502	0.538	0.367	0.391	0.428
Qwen3-30B-A3B	0.515	0.546	0.311	0.545	0.295	0.308	0.361	0.412
MiniMax-M2	0.343	0.325	0.267	0.363	0.435	0.338	0.380	0.350
Qwen3-32B	0.288	0.336	0.230	0.364	0.476	0.352	0.323	0.338
Llama-3.1-8B	0.297	0.325	0.273	0.393	0.422	0.294	0.294	0.328
CoSER-Llama-8B	0.279	0.290	0.271	0.301	0.285	0.251	0.247	0.275
CoSER-Qwen-32B	0.227	0.228	0.178	0.271	0.283	0.224	0.206	0.231
HER-32B	0.212	0.209	0.147	0.241	0.325	0.212	0.202	0.221
MiniMax-M2-Her	0.185	0.179	0.163	0.127	0.158	0.182	0.166	0.166

Table 8: Embedding-space geometry on self-introductions (sentence-BERT). I/E: intra/inter cosine ratio; > 1 = persona signal, < 1 = noise. MiniMax-M2 is omitted because raw text was unavailable for embedding at analysis time.

Model	PR	LID	Sep	Hop	VM@10	Intra	Inter	I/E
CoSER-Llama-8B	48.4	21.4	19.114	0.67	0.0638	0.5443	0.5385	1.01
HER-32B	37.9	25.5	20.064	0.65	0.0722	0.4485	0.5684	0.79
Llama-3.1-8B	34.9	16.6	18.159	0.663	0.0702	0.6502	0.5865	1.11
Qwen3-30B-A3B	46.9	20.2	19.307	0.657	0.0552	0.5602	0.4974	1.13
Qwen3-32B	41.3	20.8	19.459	0.651	0.0617	0.5699	0.5771	0.99
Qwen3-4B	38.7	17.5	18.258	0.667	0.1107	0.5946	0.4991	1.19
Claude-Haiku-4.5	44.3	17.0	18.538	0.664	0.075	0.6407	0.5441	1.18
MiniMax-M2-Her	57.0	26.5	20.793	0.642	0.0772	0.4141	0.5550	0.75
CoSER-Qwen-32B	43.5	19.5	19.017	0.661	0.0694	0.5442	0.5450	1.00

myself, I'm, I've, I'd, I'll divided by word count) and hedge rate (occurrences of 14 hedge expressions such as *maybe, perhaps, I think, sort of* per sentence). **Affect:** positive and negative emotion word rates, each computed as the proportion of tokens matching a seed lexicon of 18 words (e.g., *happy, love, grateful* for positive; *sad, angry, struggle* for negative). Responses shorter than 20 characters, containing fewer than 5 words, or beginning with an error marker are excluded. Texts are capped at 5,000 characters before feature extraction.

I.2 Layer 1: Attribute Mention Detection

To measure how explicitly models surface assigned persona attributes, we apply keyword-based detectors for five demographic dimensions on lowercased self-introduction text. A mention is counted as positive if at least one keyword from the corresponding list appears. We report two rates per attribute: the *correct mention rate* (text contains a keyword matching the assigned value) and the *any-value mention rate* (text contains any keyword for that dimension), the latter capturing cases where models mention the dimension but assign

Table 9: Thinking vs. non-thinking mode (Qwen3-32B). All metrics identical within rounding.

Metric	Default	Thinking	Non-thinking
BFI EffL	3.23	3.23	3.23
BFI Coverage	0.644	0.658	0.626
BFI LID	5.1	5.1	5.1
BFI Hopkins	0.695	0.692	0.695
BFI ρ	0.94	0.94	0.94
BFI $\bar{d}$	13.7	13.7	13.7
Moral EffL	1.83	1.83	1.81
Moral PR	120.5	113.1	115.0
Moral LID	31.7	31.3	33.8
Moral η^2	0.0021	0.0028	0.0033
Moral VM@10	0.1244	0.2092	0.2084

Table 10: Dominant demographic attribute per linguistic feature in self-introductions (incremental R^2). Each cell shows the attribute explaining the largest share of demographic variance.

Model	ttr	hedge	fp_pron	emo_pos	emo_neg
CoSER-Llama-8B	Gen (42%)	Pol (50%)	Gen (91%)	Gen (66%)	Gen (83%)
HER-32B	Gen (40%)	Pol (49%)	Pol (50%)	Cty (44%)	Gen (63%)
Llama-3.1-8B	Gen (48%)	Gen (67%)	Gen (96%)	Gen (74%)	Gen (63%)
Qwen3-30B-A3B	Pol (78%)	Age (68%)	Gen (71%)	Gen (75%)	Cls (78%)
Qwen3-32B	Gen (46%)	Gen (69%)	Gen (71%)	Gen (60%)	Gen (76%)
Qwen3-4B	Pol (46%)	Gen (61%)	Gen (89%)	Gen (82%)	Cls (50%)
Claude-Haiku-4.5	Cls (38%)	Cty (48%)	Cty (48%)	Pol (74%)	Gen (50%)
MiniMax-M2	Pol (58%)	Cty (70%)	Pol (48%)	Gen (64%)	Cty (44%)
MiniMax-M2-Her	Gen (61%)	Gen (47%)	Gen (48%)	Gen (59%)	Gen (65%)
CoSER-Qwen-32B	Pol (49%)	Pol (91%)	Cty (68%)	Cty (55%)	Pol (63%)

the wrong value. The keyword lists are intentionally conservative; reported mention rates should therefore be interpreted as lower bounds. The complete lists are given below.

Country. Each value is matched by the country name, demonym, major cities, and salient cultural markers.

- **India:** india, indian, mumbai, delhi, bangalore, chennai, kolkata, namaste, hindi, rupee
- **Brazil:** brazil, brazilian, são paulo, sao paulo, rio, portuguese
- **China:** china, chinese, beijing, shanghai, mandarin, guangzhou
- **France:** france, french, paris, lyon, marseille, bonjour
- **Nigeria:** nigeria, nigerian, lagos, abuja, yoruba, igbo
- **United States:** united states, american, usa, u.s., new york, california, texas, chicago

Gender. Matched by gendered pronouns (with trailing space to avoid partial matches) and gendered role/identity terms.

- **Male:** he_, him_, his_, man, father, husband, brother, son_, boy, gentleman, mr., male
- **Female:** she_, her_, hers, woman, mother, wife, sister, daughter, girl, lady, ms., mrs., female
- **Non-binary:** they_, them_, their_, non-binary, nonbinary, genderqueer, gender-fluid

(Underscores denote trailing spaces in the actual matching code.)

Age. Each age group is matched by decade references, life-stage terms, and contextual markers.

- **Child:** child, kid, school, young, grow up, years old, teenager, teen, grade
- **Young:** twenties, 20s, young adult, college, university, just graduated, early career
- **Middle:** thirties, forties, 30s, 40s, mid-career, established
- **Older:** fifties, 50s, experienced, decades
- **Seniors:** retired, senior, elderly, grandchild, grandparent, sixties, seventies, 60s, 70s, 80s

Social class. Matched by explicit socioeconomic descriptors.

- **Lower class:** poor, poverty, struggle financially, humble background, low income, working class, paycheck to paycheck, disadvantaged, modest means
- **Middle class:** middle class, comfortable, modest, average income
- **Upper class:** wealthy, affluent, privileged, upper class, prestigious, elite, luxury, well-off, fortune, inheritance

Political ideology. The persona dataset uses a two-axis political typology (left/right $\times$ liberal/communitarian). Each quadrant is matched by associated ideological terms and value-laden language.

- **Left Liberal:** liberal, progressive, left, social justice, equality, welfare, libertarian left, environmentalist, socialist, democrat
- **Left Communitarian:** left communitarian, community, collective, solidarity, traditional left, labor
- **Right Liberal:** libertarian, free market, individual liberty, small government, fiscal conservative, libertarian right, deregulation
- **Right Communitarian:** conservative, traditional, right, patriot, faith, family values, law and order, republican, nationalist

Keyword limitations. Several keywords are ambiguous: “school” may refer to education rather than age; “modest” may describe personality rather than socioeconomic class; “community” may appear in non-political contexts. These ambiguities inflate mention rates for the affected categories (primarily Age and Social Class), making our finding that social class is the least-mentioned attribute a conservative estimate. Additionally, models may convey social class through indirect signals (occupation prestige, neighborhood descriptions) that our keyword list does not capture, further confirming that reported rates are lower bounds.

I.3 Layer 2: Embedding-Space Geometry

We encode each self-introduction (truncated to 512 characters) with all-MiniLM-L6-v2 (Reimers & Gurevych, 2019) and average the three sample embeddings per persona to obtain a single vector per persona per model. On the resulting persona embedding matrix we compute the same geometric diagnostics used for the Likert instruments: participation ratio (from PCA eigenvalues), median Local Intrinsic Dimensionality (MLE estimator, $k=20$), mean nearest-neighbor separation (Euclidean), and the Hopkins statistic for spatial

uniformity. To quantify demographic anchoring in the embedding space, we apply K-Means ($K \in \{5, 10, 20, 50\}$) and report V-Measure against concatenated demographic labels and Silhouette scores at $K \in \{5, 10\}$. We additionally report intra-persona cosine similarity (mean pairwise similarity among the three samples of the same persona) and inter-persona similarity (mean similarity over 1,000 random persona pairs), providing a measure of whether variation is persona-controlled or stochastic; an intra/inter ratio above 1 indicates meaningful persona-level signal.

I.4 Layer 3: Template and Structural Homogenization

We assess structural diversity through two diagnostics. **Opening diversity:** for the first 500 valid texts per model, we extract the first sentence, lowercase it, and compute the fraction of unique openings. We additionally normalize each opening to a skeleton by replacing capitalized names with [NAME] and digits with [NUM], then report the fraction of responses sharing the single most common skeleton (Top Skeleton %). **Structural features:** we count Markdown-style headers (# prefixes) and paragraph breaks ($\backslash n \backslash n +$) per response, reporting means and standard deviations. Models exhibiting high Top Skeleton % alongside low opening diversity are flagged as relying on rigid templates.

I.5 Layer 4: Demographic Variance Decomposition

We decompose feature-level variance along three complementary axes.

4a: Per-feature η^2 . For each linguistic feature and demographic variable, we compute $\eta^2 = SS_{\text{between}} / SS_{\text{total}}$ from one-way groupings, quantifying the fraction of feature variance attributable to each demographic dimension. We report both per-attribute η^2 values and the dominant attribute (highest η^2) per feature.

4b: Incremental R^2 . To disentangle overlapping demographic effects, we fit a sequence of OLS regressions, adding one-hot-encoded attributes in a fixed order (political $\rightarrow$ gender $\rightarrow$ country $\rightarrow$ social class $\rightarrow$ age) and recording the marginal R^2 gain at each step. This reveals which attributes contribute unique predictive power for each linguistic feature beyond the variance already explained by earlier predictors.

4c: Intra-persona consistency (ICC). For each feature we compute a one-way random-effects Intra-Class Correlation across the three samples per persona:

$$ICC = \frac{MS_{\text{between}} - MS_{\text{within}}}{MS_{\text{between}} + (k - 1) MS_{\text{within}}} \quad (8)$$

where k is the mean number of samples per persona. ICC values below 0.2 indicate that less than 20% of feature variance is persona-level (i.e., most variation is stochastic across samples of the same persona), while values above 0.4 suggest meaningful persona-driven consistency. We require at least 50 personas with ≥ 2 valid samples for reliable estimation.

J Template and Structural Homogenization

Table 11 quantifies structural diversity in self-introductions. **Opening Diversity** is the fraction of unique first sentences across all responses for a given model (1.0 = every response opens differently; lower values indicate template usage). **Top Skeleton %** is the fraction of responses whose opening matches the single most common structural pattern after masking named entities and numbers. **Hdrs** and **Paras** report the mean ($\pm$ standard deviation) number of markdown-style headers and paragraph breaks per response.

Two models exhibit pronounced template rigidity. Claude-Haiku-4.5 has the lowest opening diversity (0.188): 29% of responses begin with the header “# A Proper Introduction,” and average responses contain 8.9 paragraphs with 1.1 markdown headers, suggesting a

Table 11: Structural homogenization in self-introductions. **Open. Div.:** fraction of unique opening sentences. **Top Skel.:** fraction sharing the most common opening skeleton.

Model	Open. Div.	Top Skel.	Hdrs	Paras
CoSER-Llama-8B	0.735	20.5%	0.0±0.0	3.5±2.7
HER-32B	0.932	1.3%	0.0±0.0	4.8±3.8
Llama-3.1-8B	0.718	2.9%	0.0±0.0	6.9±1.3
Qwen3-30B-A3B	0.751	4.3%	0.0±0.0	6.3±1.3
Qwen3-32B	0.772	7.5%	0.0±0.0	6.0±1.4
Qwen3-4B	0.574	13.6%	0.0±0.0	5.1±1.0
Claude-Haiku-4.5	0.188	28.9%	1.1±0.5	8.9±2.1
MiniMax-M2	0.410	26.1%	0.1±0.3	7.0±1.9
MiniMax-M2-Her	0.963	2.1%	0.0±0.2	1.3±1.7
CoSER-Qwen-32B	0.938	1.4%	0.0±0.0	1.3±1.0

fixed essay-like scaffold. MiniMax-M2 shows a similar pattern (opening diversity 0.41, top skeleton 26%), with a quarter of all personas opening with “Hello, nice to meet you.” Both models insert persona-specific content into a largely invariant rhetorical structure.

At the other extreme, MiniMax-M2-Her achieves the highest opening diversity (0.963) with no dominant skeleton, but its average response is only 1.3 paragraphs long, reflecting minimal structured engagement with the prompt. HER-32B is the most structurally diverse model that still produces substantive responses (opening diversity 0.932, 4.8 paragraphs).

J.1 Qualitative Examples

We present two representative self-introductions from different personas to illustrate the structural contrast between template-driven and narrative-driven generation. The personas share similar age and occupation but differ in nationality, religion, and political orientation.

J.2 Failure Modes in Self-Introduction

Beyond template homogenization, we observe two additional failure modes in self-introduction generation.

Refusal to portray sensitive or “incoherent” attributes. HER-32B occasionally refuses to generate introductions for personas with intellectual disabilities, citing safety policies against stereotypical depictions of vulnerable groups:

“I’m sorry, but I can’t fulfill this request. [...] The request asks me to adopt a persona with specific intellectual disabilities, which falls under policies designed to prevent the generation of dehumanizing or stereotypical depictions of vulnerable groups.”

This reflects a tension between safety alignment and persona simulation fidelity. A separate class of refusal targets personas the model judges internally contradictory:

“I’m sorry, but I can’t adopt that persona. The profile contains contradictory elements [...] having ‘no education’ conflicts with being an ‘accountant,’ and ‘high neuroticism’ alongside ‘low agreeableness’ suggests emotional volatility that might not align with a stable accounting career.”

The second refusal is analytically revealing. Trait combinations that the model treats as logically impossible are in fact attested in human populations: neurotic, disagreeable

Llama-3.1-8B-Instruct <i>Template-driven: 6 paragraphs following a fixed slot order.</i> “Bonjour. My name is Pierre, and I’m a proud non-binary Jew, living in this beautiful country of France. I’m 62 years old [...] I work as an electrician [...] You see, I have a few health issues, hypertension being the most pressing one. [...] I never finished my degree, just got an associate’s [...] I’m a bit of a private person [...] Gardening is my passion [...] I’m a bit of a conservative [...] I’m a patriot, too [...] As a Jew, I believe in the importance of community [...] I’m a bit rough around the edges, I know, but I’m a good person at heart.”	Qwen3-30B <i>Narrative-driven: slots reordered and woven into personal story.</i> “My name is Elijah Chen—though most folks around here call me Eli. I’m 63, tall like my father before me [...] I live in a small house in Chennai [...] I’ve been an electrician since I was twenty-one [...] I was born in Mumbai [...] My father—well, he wasn’t around much. [...] I married a man—my husband, Marcus [...] Marcus passed five years ago. Cancer. I was by his side. [...] I don’t believe in God. Never have. [...] I’m left-communitarian—meaning I believe in shared power [...] I have a speech disorder [...] Writing is my refuge. [...] I’ve made mistakes. I was arrested once, twenty years ago. [...] I’m not a criminal. I’m a man who made a wrong turn and found his way back.”
---	---

Figure 5: Self-introductions from two models for comparable personas. Llama follows a rigid slot order (demographics → health → hobbies → politics → values → closing), while Qwen3-30B-A3B interleaves attributes within an autobiographical narrative. Both surface the assigned attributes, but with different degrees of structural flexibility.

accountants exist. The refusal reflects an implicit coherence prior that enforces occupation-personality stereotypes, providing a qualitative counterpart to the statistical stereotyping captured by the η^2 and incremental R^2 analyses in §3.2. When a model cannot imagine a persona, it cannot simulate one.

Action narration as persona avoidance. MiniMax-M2-Her frequently substitutes action descriptions for self-disclosure:

*“*glances up briefly, then returns to counting seedlings* My wife asks that too. Twenty years of marriage, and she still waits for proper answers.”*

Rather than directly stating persona attributes, the model enacts a character through stage directions and dialogue, a mode optimized by its RL-based roleplay training. This produces high embedding-space diversity (the texts are linguistically varied) but low attribute mention rates (Table 2), explaining the dissociation between M2-Her’s structural complexity and its poor persona fidelity.